Last name _____

First name _____

LARSON—MATH 310–HOMEWORK WORKSHEET 09
QR Factorization

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Gram-Schmidt algorithm for k vectors.

Input: $\vec{a}_1, \vec{a}_2, \ldots \vec{a}_k$.

Output: on each step get a $\vec{0}$ vector, which is necessarily a linear combination of the previously created $\vec{q}$'s **or** a new $\vec{q}$. The produced $\vec{q}$ vectors are orthonormal.

1. Let $\vec{q}_1 = \frac{1}{||\vec{a}_1||}\vec{a}_1$. (So $\vec{a}_1 = ||\vec{a}_1|| \cdot \vec{q}_1$ This means $\vec{a}_1$ is a linear combination of the first 1 $\vec{q}$'s).
2. Let $\tilde{q}_2 = \vec{a}_2 - (\vec{q}_1^T \vec{a}_2)\vec{q}_1$. (If $\tilde{q}_2 = \vec{0}$ then $\vec{a}_2 = (\vec{q}_1^T \vec{a}_2)\vec{q}_1$. Try the next $\vec{a}$ to produce $\vec{q}_2$).
3. Let $\vec{q}_2 = \frac{1}{||\tilde{q}_2||}\tilde{q}_2$. (So $\vec{a}_2 = ||\tilde{q}_2|| \cdot \vec{q}_2 + (\vec{q}_1^T \vec{a}_2)\vec{q}_1$. This means $\vec{a}_2$ is a linear combination of the first 2 $\vec{q}$'s).
4. Let $\tilde{q}_3 = \vec{a}_3 - (\vec{q}_1^T \vec{a}_3)\vec{q}_1 - (\vec{q}_2^T \vec{a}_3)\vec{q}_2$. (If $\tilde{q}_3 = \vec{0}$ then $\vec{a}_3 = (\vec{q}_1^T \vec{a}_3)\vec{q}_1 - (\vec{q}_2^T \vec{a}_3)\vec{q}_2$. Try the next $\vec{a}$ to produce $\vec{q}_3$).
5. Let $\vec{q}_3 = \frac{1}{||\tilde{q}_3||}\tilde{q}_3$. (So $\vec{a}_3 = ||\tilde{q}_3|| \cdot \vec{q}_3 + (\vec{q}_1^T \vec{a}_2)\vec{q}_1 + (\vec{q}_2^T \vec{a}_3)\vec{q}_2$. This means $\vec{a}_3$ is a linear combination of the first 3 $\vec{q}$'s).
6. Keep going. At each iteration, you either get a $\vec{0}$ vector and the currently processed $\vec{a}$ vector is a linear combination of the previously existing $\vec{q}$ vectors **or** the current $\vec{a}$ vector yields a new orthogonal $\vec{q}$ vector, and the currently processed $\vec{a}$ vector is a linear combination of the produced $\vec{q}$ vectors.

QR Factorization

1. Let the columns of matrix A be the $\vec{a}$ vectors.
2. Apply Gram-Schmidt to get orthogonal $\vec{q}$ vectors. Write each $\vec{a}$ as a linear combination of $\vec{q}$ vectors.
3. Let Q be a matrix whose columns are the produced $\vec{q}$ vectors.

4. Let R be the matrix with the same number of columns as A and whose i^{th} column is the coefficients of $\vec{a}_i$ written as a linear combination of $\vec{q}$ vectors. (The Gram-Schmidt algorithm guarantees that $veca_i$ must be written with $vecq$ vectors whose indices are no more than i, and so the coefficients of the $\vec{q}$ vectors with index greater than i are 0. So R **must** have 0's below its diagonal.

5. Then $A = QR$.

Problems

Let the columns of matrix A be $\vec{a}_1, \ldots \vec{a}_6$.

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. (a) Apply Gram-Schmidt to find an orthonormal list of $\vec{q}$ vectors.
   (b) Use these as columns for the Q matrix.
   (c) Check that $QQ^T = I$.
   (d) Write each $\vec{a}$ vector as a linear combination of the $\vec{q}$ vectors.
   (e) Use the coefficients from the last problem as the columns for your R matrix.
   (f) Check that $A = QR$. If not, go back and find your error.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 3 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 5 & 5 \end{bmatrix}$$

2. (a) Apply Gram-Schmidt to find an orthonormal list of $\vec{q}$ vectors.
   (b) Use these as columns for the Q matrix.
   (c) Check that $QQ^T = I$.
   (d) Write each $\vec{a}$ vector as a linear combination of the $\vec{q}$ vectors.
   (e) Use the coefficients from the last problem as the columns for your R matrix.
   (f) Check that $A = QR$. If not, go back and find your error.